# Linked List Program

## 1. Singly Linked List

**Q. 1 Write a C program to create a singly linked list.**

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
   int data;
   struct node *next;
};
struct node *p=NULL,*q;
void main()
{
   int d;
   int choice;
   do
   {
         if(p==NULL)
         {
                 p=(struct node *)malloc(sizeof(struct node));
                 q=p;
         }
         else
         {
                q->next=(struct node *)malloc(sizeof(struct node));
                q=q->next;
         }
         printf("\nEnter your value to insert into node=");
         scanf("%d",&d);
         q->data=d;
         q->next=NULL;
         printf("\nDo you want to add one more node( Yes=1/No=0)");
         scanf("%d",&choice);
   }while(choice==1);
   q=p;
   while(q!=NULL)
   {
     printf("\t%d",q->data);
     q=q->next;
   }
}
```

## Insertion into Singly Linked List

**Q.1 Write a program in C to insert a node at the beginning**

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
   int data;
   struct node *next;
};
struct node *p=NULL,*q,*tmp;
void main()
{
   int d;
   int choice;
   do
   {
         if(p==NULL)
         {
               p=(struct node *)malloc(sizeof(struct node));
               q=p;
         }
         else
         {
               q->next=(struct node *)malloc(sizeof(struct node));
               q=q->next;
         }
         printf("\nEnter your value to insert into node=");
         scanf("%d",&d);
         q->data=d;
         q->next=NULL;
         printf("\nDo you want to add one more node( Yes=1/No=0)");
         scanf("%d",&choice);
   }while(choice==1);
   q=p;
   while(q!=NULL)
   {
     printf("\t%d",q->data);
     q=q->next;
   }
   q=p;
   printf("\nEnter data to be insert at begining=");
   scanf("%d",&d);
   tmp=(struct node *)malloc(sizeof(struct node));
   tmp->data=d;
   tmp->next=p;
   p=tmp;
   printf("List after inserting node at begining=");
   q=p;
   while(q!=NULL)
```

```
    {
        printf(" %d",q->data);
        q=q->next;
    }
}
```

**Q.2 Write a program in c to insert a node at the ending**

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *p=NULL,*q,*tmp,*last;
void main()
{
    int d;
    int choice;
    do
    {
            if(p==NULL)
            {
                    p=(struct node *)malloc(sizeof(struct node));
                    q=p;
            }
            else
            {
                    q->next=(struct node *)malloc(sizeof(struct node));
                    q=q->next;
            }
            printf("\nEnter your value to insert into node=");
            scanf("%d",&d);
            q->data=d;
            q->next=NULL;
            printf("\nDo you want to add one more node( Yes=1/No=0)");
            scanf("%d",&choice);
    }while(choice==1);
    q=p;
    while(q!=NULL)
    {
      last=q;
      printf("\t%d",q->data);
      q=q->next;
    }
    printf("\nEnter data to be insert at ending=");
    scanf("%d",&d);
    tmp=(struct node *)malloc(sizeof(struct node));
    tmp->data=d;
    tmp->next=NULL;
    last->next=tmp;
```

```
    printf("List after inserting node at ending=");
    q=p;
    while(q!=NULL)
    {
        printf(" %d",q->data);
        q=q->next;
    }
   }
```

**Q.3 Write a program in c to insert a node at a given position.**

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *p=NULL,*q,*tmp,*last;
void main()
{
    int d,pos;
    int choice,count=0;
    do
    {
        if(p==NULL)
        {
            p=(struct node *)malloc(sizeof(struct node));
            q=p;
        }
        else
        {
            q->next=(struct node *)malloc(sizeof(struct node));
            q=q->next;
        }
        printf("\nEnter your value to insert into node=");
        scanf("%d",&d);
        q->data=d;
        q->next=NULL;
        printf("\nDo you want to add one more node( Yes=1/No=0)");
        scanf("%d",&choice);
    }while(choice==1);
    q=p;
    while(q!=NULL)
    {
      last=q;
      printf("\t%d",q->data);
      q=q->next;
    }
    printf("\nEnter your position where you want to insert node=");
    scanf("%d",&pos);
    q=p;
```

```
    while(q!=NULL)
    {
        if(count==pos)
        {
            printf("\nEnter your node data to insert at %d position",pos);
            scanf("%d",&d);
            tmp=(struct node *)malloc(sizeof(struct node));
            tmp->data=d;
            tmp->next=q->next;
            q->next=tmp;
        }
        q=q->next;
        count++;
    }
   q=p;
   printf("\nFinal List=");
   while(q!=NULL)
    {
      last=q;
      printf("\t%d",q->data);
      q=q->next;
    }

}
```

**Searching operation on Singly Linked List**

**Q.1 Write a C program to implement linear search on a singly linked list.**

Q 5. Write a program in c to remove the element from the singly linked list.

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *p=NULL,*q,*tmp,*last;
void main()
{
    int d,pos;
    int choice,count=0,target;
    do
    {
        if(p==NULL)
        {
            p=(struct node *)malloc(sizeof(struct node));
            q=p;
```

```
            }
            else
            {
                        q->next=(struct node *)malloc(sizeof(struct node));
                        q=q->next;
            }
            printf("\nEnter your value to insert into node=");
            scanf("%d",&d);
            q->data=d;
            q->next=NULL;
            printf("\nDo you want to add one more node( Yes=1/No=0)");
            scanf("%d",&choice);
   }while(choice==1);
   q=p;
   while(q!=NULL)
   {
     last=q;
     printf("\t%d",q->data);
     q=q->next;
   }
   printf("\nEnter your node target data to remove the node=");
   scanf("%d",&target);
   q=p;
   while(q->next!=NULL)
   {
      if(q->next->data==target)
      {
        tmp=q->next;
        q->next=tmp->next;
        free(tmp);
      }
      q=q->next;
      count++;
   }
  q=p;
  printf("\nFinal List=");
  while(q!=NULL)
   {
     last=q;
     printf("\t%d",q->data);
     q=q->next;
   }

}
```

**Q. Write a C program to implement a doubly linked list.**

```
#include<stdio.h>
#include<stdlib.h>
struct dnode
{
    struct dnode *prev;
    int data;
    struct dnode *next;
};
struct dnode *p=NULL,*q, *last;
void main()
{
    int d;
    int choice;
    do
    {
            if(p==NULL)
            {
                    p=(struct dnode *)malloc(sizeof(struct dnode));
                    p->prev=NULL;
                    q=p;
            }
            else
            {
                   q->next=(struct dnode *)malloc(sizeof(struct dnode));
                    q->next->prev=q;
                   q=q->next;
            }
            printf("\nEnter your value to insert into node=");
            scanf("%d",&d);
            q->data=d;
            q->next=NULL;
            printf("\nDo you want to add one more node( Yes=1/No=0)");
            scanf("%d",&choice);
    }while(choice==1);
    q=p;
    while(q!=NULL)
    {
      last=q;
      printf("\t%d",q->data);
```

```
      q=q->next;
    }
    q=last;
    printf("\n");
    while(q!=NULL)
    {
      printf("\t%d",q->data);
      q=q->prev;
    }
}
```

**//Linear Search on Singly Linked List**

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *p=NULL,*q;
void main()
{
    int d,target,choice,position=1,flag=0;
    do
    {
            if(p==NULL)
            {
                    p=(struct node *)malloc(sizeof(struct node));
                    q=p;
            }
            else
            {
                   q->next=(struct node *)malloc(sizeof(struct node));
                   q=q->next;
            }
            printf("\nEnter your value to insert into node=");
            scanf("%d",&d);
            q->data=d;
            q->next=NULL;
            printf("\nDo you want to add one more node( Yes=1/No=0)");
            scanf("%d",&choice);
    }while(choice==1);
    q=p;
    while(q!=NULL)
    {
      printf("\t%d",q->data);
      q=q->next;
    }
    printf("\nEnter your target value to search");
    scanf("%d",&target);
    q=p;
```

```
    while(q!=NULL)
    {
        if(q->data==target)
        {
            printf("Element %d found at %d position ",target,position);
            flag=1;break;
        }
        q=q->next;
        position++;
    }
    if(flag==0)
    {
            printf("Element %d not found",target);
    }
}
```